

6.4.4 Implementation Process

6.4.4.1 Purpose

The purpose of the Implementation Process is to realize a specified system element in accordance with its design and requirements.

NOTE: Users of this International Standard intending to implement software products or software elements of a larger system shall apply the **Software Implementation Process** (subclause 7.1.1), which is a conforming instance of the Implementation Process.

6.4.4.2 Outcomes

As a result of successful implementation of the Implementation Process:

- a) An implementation strategy is defined that aligns with system design and applicable standards.
 - b) The system element is realized in accordance with its specified design and requirements.
 - c) The implemented element is verified against its allocated requirements.
 - d) The implemented element is packaged, stored, and made available for subsequent integration, testing, or deployment, with configuration status recorded.
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6.4.4.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.4.3.1 Implementation Planning

- 6.4.4.3.1.1 Define and document the implementation strategy, including applicable standards, methods, tools, environments, and responsibilities.
- 6.4.4.3.1.2 Identify system elements to be implemented, referencing System Architectural Design (6.4.3) for interfaces and dependencies.
- 6.4.4.3.1.3 Establish verification and acceptance criteria for the implemented element.

6.4.4.3.2 Implementation Execution

- 6.4.4.3.2.1 Realize the system element according to the defined strategy, applying required standards, methods, and tools.
- 6.4.4.3.2.2 Maintain traceability between implementation artifacts, requirements, and design documentation.
- 6.4.4.3.2.3 Apply configuration management controls (6.3.4) to all implementation baselines.

6.4.4.3.3 Verification and Packaging

- 6.4.4.3.3.1 Verify the implemented element against allocated requirements using appropriate methods (inspection, analysis, demonstration, or test).
- 6.4.4.3.3.2 Record verification results and handle nonconformances per organizational procedures.
- 6.4.4.3.3.3 Package, store, and deliver the verified element for integration, ensuring documentation and configuration status accounting are complete.

6.4.5 System Integration Process

6.4.5.1 Purpose

The purpose of the System Integration Process is to integrate system elements—including software items, hardware items, manual operations, and other systems—into a complete system that satisfies the system design and customer expectations expressed in the system requirements.

6.4.5.2 Outcomes

As a result of successful implementation of the System Integration Process:

- a) A strategy to integrate the system according to the priorities of the system requirements is developed.
- b) Criteria are defined to verify compliance with system requirements allocated to system elements, including interfaces between system elements.

- c) System integration is verified using the defined criteria.
 - d) A regression strategy is developed and applied for re-testing the system when changes are made.
 - e) Consistency and traceability are established between the system design and the integrated system elements.
 - f) An integrated system is constructed that demonstrates compliance with the system design and includes a complete set of usable deliverable system elements.
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6.4.5.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.5.3.1 Integration Planning

This activity establishes the strategy, criteria, and sequence for system integration.

- 6.4.5.3.1.1 Define and document an integration strategy that prioritizes system requirements and establishes the integration sequence of system elements.
 - 6.4.5.3.1.2 Define integration interfaces, responsibilities, tools, and methods.
 - 6.4.5.3.1.3 Establish criteria and procedures for integration verification.
 - 6.4.5.3.1.4 Ensure consistency with System Architectural Design (6.4.3) and Configuration Management (6.3.4).
 - 6.4.5.3.1.5 Plan for regression testing when changes occur.
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6.4.5.3.2 System Integration

This activity integrates elements and verifies the integrated aggregates.

- 6.4.5.3.2.1 Integrate software configuration items, hardware configuration items, manual operations, and other systems as necessary into the system according to the integration strategy.
- 6.4.5.3.2.2 Test integrated aggregates against their allocated requirements as integration progresses.
- 6.4.5.3.2.3 Document integration steps and test results to ensure traceability.
- 6.4.5.3.2.4 Address consistency and traceability between the system

design and the integrated system configuration.

6.4.5.3.3 Test Readiness

This activity ensures the integrated system is ready for qualification testing.

- 6.4.5.3.3.1 Develop and document tests, test cases, criteria, and procedures for each qualification requirement of the system.
- 6.4.5.3.3.2 Evaluate the readiness of the integrated system for System Qualification Testing (6.4.6) using the following criteria:
 - Test coverage of system requirements
 - Appropriateness of test methods and standards
 - Conformance to expected results
 - Feasibility of executing system qualification testing
- 6.4.5.3.3.3 Develop and maintain a regression testing strategy to support re-testing after changes.
- 6.4.5.3.3.4 Record and report test readiness evaluations to relevant stakeholders.